

Parallel Computing – SE3082

Lab 07

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Exercise 1

```
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercisel.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex1 exercisel.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex1
Process 1: sum 2500001 to 5000001 = 9375006250001
Process 3: sum 7500001 to 10000000 = 21875001250000
Process 2: sum 5000001 to 7500001 = 15625008750001
Process 0: sum 1 to 2500001 = 3125003750001
Total sum 1 to 10000000 = 50000020000003
```

Exercise 2

```
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano create_numbers.py
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ python3 create_numbers.py
Created create_numbers.csv with 10,000 random numbers in vertical format
Each number is on a separate line for easy viewing in Excel
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ |
```

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geeth@DESKTOP-3400SVB: /  ×  +  ▾

GNU nano 7.2 create_numbers.py *
import random

# Create a CSV file with 10,000 random numbers in vertical format
with open('numbers.csv', 'w') as f:
    for i in range(10000):
        f.write(f"{random.randint(1, 1000)}")
        if i < 9999:
            f.write("\n")

print("Created create_numbers.csv with 10,000 random numbers in vertical format")
print("Each number is on a separate line for easy viewing in Excel")
```

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format.

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A1									
	A	B	C	D	E	F	G	H	I
9978	147								
9979	572								
9980	284								
9981	780								
9982	49								
9983	449								
9984	106								
9985	174								
9986	820								
9987	426								
9988	438								
9989	796								
9990	47								
9991	36								
9992	184								
9993	728								
9994	152								
9995	83								
9996	189								
9997	750								
9998	104								
9999	244								
10000	124								
10001									

```

geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise2.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex2 exercise2.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex2
Root: Read 10000 numbers from file
Process 0: Received broadcast data
Process 0: Partial sum = 1246202.00
Final total sum: 5021552.00
Process 2: Received broadcast data
Process 2: Partial sum = 1267921.00
Process 3: Received broadcast data
Process 3: Partial sum = 1257201.00
Process 1: Received broadcast data
Process 1: Partial sum = 1250228.00

```

Exercise 3

```

geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise3.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex3 exercise3.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex3
Process 1: Received scattered data
Process 1: Partial sum = 1250228.00
Process 2: Received scattered data
Process 2: Partial sum = 1267921.00
Process 3: Received scattered data
Process 3: Partial sum = 1257201.00
Root: Read 10000 numbers from file
Process 0: Received scattered data
Process 0: Partial sum = 1246202.00
Final total sum: 5021552.00

```

Exercise 4

```

geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise4.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex4 exercise4.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex4
Root: Read 10000 numbers from file
Process 1: Received scattered data
Process 1: Partial sum = 1250228.00
Process 2: Received scattered data
Process 2: Partial sum = 1267921.00
Process 0: Received scattered data
Process 3: Received scattered data
Process 0: Partial sum = 1246202.00
Partial sum from process 0: 1246202.00
Partial sum from process 1: 1250228.00
Partial sum from process 2: 1267921.00
Partial sum from process 3: 1257201.00
Final total sum: 5021552.00
Process 3: Partial sum = 1257201.00

```

Exercise 5

```

geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise5.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex5 exercise5.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex5
Root: Read 10000 numbers from file
Process 0: Received scattered data
Process 0: Partial sum = 1246202.00
Final total sum: 5021552.00
Process 1: Received scattered data
Process 1: Partial sum = 1250228.00
Process 2: Received scattered data
Process 2: Partial sum = 1267921.00
Process 3: Received scattered data
Process 3: Partial sum = 1257201.00

```

Exercise 6

```
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise6.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex6 exercise6.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 4 ./ex6
Process 3: Received scattered data
Process 3: Partial sum = 1257201.00
Process 3: Total sum = 5021552.00
Root: Read 10000 numbers from file
Process 0: Received scattered data
Process 0: Partial sum = 1246202.00
Process 0: Total sum = 5021552.00
Process 1: Received scattered data
Process 1: Partial sum = 1250228.00
Process 1: Total sum = 5021552.00
Process 2: Received scattered data
Process 2: Partial sum = 1267921.00
Process 2: Total sum = 5021552.00
```

Exercise 7

```
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise7.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex7 exercise7.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpiexec -n 2 ./ex7
=== MPI Ping-Pong Array Growth Program ===
Starting ping-pong with 100 message exchanges

[Master] Exchange 1: Sending array of length 1 (added 55)
[Worker] Exchange 1: Received array of length 1
[Worker] Exchange 1: Sending back array of length 2 (added 863)
[Worker] Exchange 2: Received array of length 3
[Worker] Exchange 2: Sending back array of length 4 (added 146)
[Worker] Exchange 3: Received array of length 5
[Worker] Exchange 3: Sending back array of length 6 (added 296)
[Worker] Exchange 4: Received array of length 7
[Worker] Exchange 4: Sending back array of length 8 (added 152)
[Worker] Exchange 5: Received array of length 9
[Worker] ... (continuing) ...
[Master] Exchange 1: Received array of length 2
[Master] Exchange 2: Sending array of length 3 (added 705)
[Master] Exchange 2: Received array of length 4
[Master] Exchange 3: Sending array of length 5 (added 38)
[Master] Exchange 3: Received array of length 6
[Master] Exchange 4: Sending array of length 7 (added 177)
[Master] Exchange 4: Received array of length 8
[Master] Exchange 5: Sending array of length 9 (added 872)
[Master] ... (continuing) ...
[Master] Exchange 95: Received array of length 190
[Master] Exchange 96: Sending array of length 191 (added 9)
[Master] Exchange 96: Received array of length 192
[Master] Exchange 97: Sending array of length 193 (added 231)
[Master] Exchange 97: Received array of length 194
[Master] Exchange 98: Sending array of length 195 (added 432)
[Master] Exchange 98: Received array of length 196
[Master] Exchange 99: Sending array of length 197 (added 832)
[Master] Exchange 99: Received array of length 198
[Master] Exchange 100: Sending array of length 199 (added 949)

=== Summary ===
Total exchanges: 100
Final array length: 199
Total time: 0.000324 seconds
```

```
[Worker] Exchange 5: Received array of length 9
[Worker] ... (continuing) ...
[Master] Exchange 1: Received array of length 2
[Master] Exchange 2: Sending array of length 3 (added 705)
[Master] Exchange 2: Received array of length 4
[Master] Exchange 3: Sending array of length 5 (added 38)
[Master] Exchange 3: Received array of length 6
[Master] Exchange 4: Sending array of length 7 (added 177)
[Master] Exchange 4: Received array of length 8
[Master] Exchange 5: Sending array of length 9 (added 872)
[Master] ... (continuing) ...
[Master] Exchange 95: Received array of length 190
[Master] Exchange 96: Sending array of length 191 (added 9)
[Master] Exchange 96: Received array of length 192
[Master] Exchange 97: Sending array of length 193 (added 231)
[Master] Exchange 97: Received array of length 194
[Master] Exchange 98: Sending array of length 195 (added 432)
[Master] Exchange 98: Received array of length 196
[Master] Exchange 99: Sending array of length 197 (added 832)
[Master] Exchange 99: Received array of length 198
[Master] Exchange 100: Sending array of length 199 (added 949)

=== Summary ===
Total exchanges: 100
Final array length: 199
Total time: 0.000324 seconds
Average time per exchange: 0.000003 seconds

First 10 elements: 55 863 705 146 38 296 177 152 872 942
Last 10 elements: 314 9 574 231 907 432 242 832 242 949
[Worker] Exchange 95: Sending back array of length 190 (added 314)
[Worker] Exchange 96: Received array of length 191
[Worker] Exchange 96: Sending back array of length 192 (added 574)
[Worker] Exchange 97: Received array of length 193
[Worker] Exchange 97: Sending back array of length 194 (added 907)
[Worker] Exchange 98: Received array of length 195
[Worker] Exchange 98: Sending back array of length 196 (added 242)
[Worker] Exchange 99: Received array of length 197
[Worker] Exchange 99: Sending back array of length 198 (added 242)
[Worker] Exchange 100: Received array of length 199
```

Exercise 8

```
^Cgeeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ nano exercise8.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpicc -o ex8 exercise8.c
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpirun -n 4 ./ex8
=== MPI Ring Topology Array Passing ===
Number of processes: 4
Number of cycles: 2
Total passes through nodes: 8

[Rank 0] Pass 1: Added rank 0 to array (length: 1)
[Rank 1] Pass 2: Added rank 1 to array (length: 2)
[Rank 2] Pass 3: Added rank 2 to array (length: 3)
[Rank 0] Pass 5: Added rank 0 to array (length: 5)

=== Ring Passing Complete ===
Total time: 0.000624 seconds
Final array length: 8

=== Final Array Contents ===
0 1 2 3 0 1 2 3

=== Verification ===
Expected pattern: Each rank appears 2 times
Rank occurrences:
  Rank 0: 2 times
  Rank 1: 2 times
  Rank 2: 2 times
  Rank 3: 2 times

All processes completed successfully.
[Rank 1] Pass 6: Added rank 1 to array (length: 6)
[Rank 3] Pass 4: Added rank 3 to array (length: 4)
[Rank 3] Pass 8: Added rank 3 to array (length: 8)
[Rank 2] Pass 7: Added rank 2 to array (length: 7)
```

```
geeth@DESKTOP-3400SVB:/mnt/c/Users/geeth2001/Desktop/3rd Year/PC/Labs/lab07$ mpirun -n 6 ./ex8
=== MPI Ring Topology Array Passing ===
Number of processes: 6
Number of cycles: 2
Total passes through nodes: 12

[Rank 0] Pass 1: Added rank 0 to array (length: 1)
[Rank 1] Pass 2: Added rank 1 to array (length: 2)
[Rank 2] Pass 3: Added rank 2 to array (length: 3)
[Rank 5] Pass 6: Added rank 5 to array (length: 6)
[Rank 3] Pass 4: Added rank 3 to array (length: 4)
[Rank 4] Pass 5: Added rank 4 to array (length: 5)
[Rank 0] Pass 7: Added rank 0 to array (length: 7)

=== Ring Passing Complete ===
Total time: 0.000287 seconds
Final array length: 12

=== Final Array Contents ===
0 1 2 3 4 5 0 1 2 3 4 5

=== Verification ===
Expected pattern: Each rank appears 2 times
Rank occurrences:
  Rank 0: 2 times
  Rank 1: 2 times
  Rank 2: 2 times
  Rank 3: 2 times
  Rank 4: 2 times
  Rank 5: 2 times
[Rank 1] Pass 8: Added rank 1 to array (length: 8)
[Rank 3] Pass 10: Added rank 3 to array (length: 10)
[Rank 2] Pass 9: Added rank 2 to array (length: 9)
[Rank 4] Pass 11: Added rank 4 to array (length: 11)

All processes completed successfully.
[Rank 5] Pass 12: Added rank 5 to array (length: 12)
```